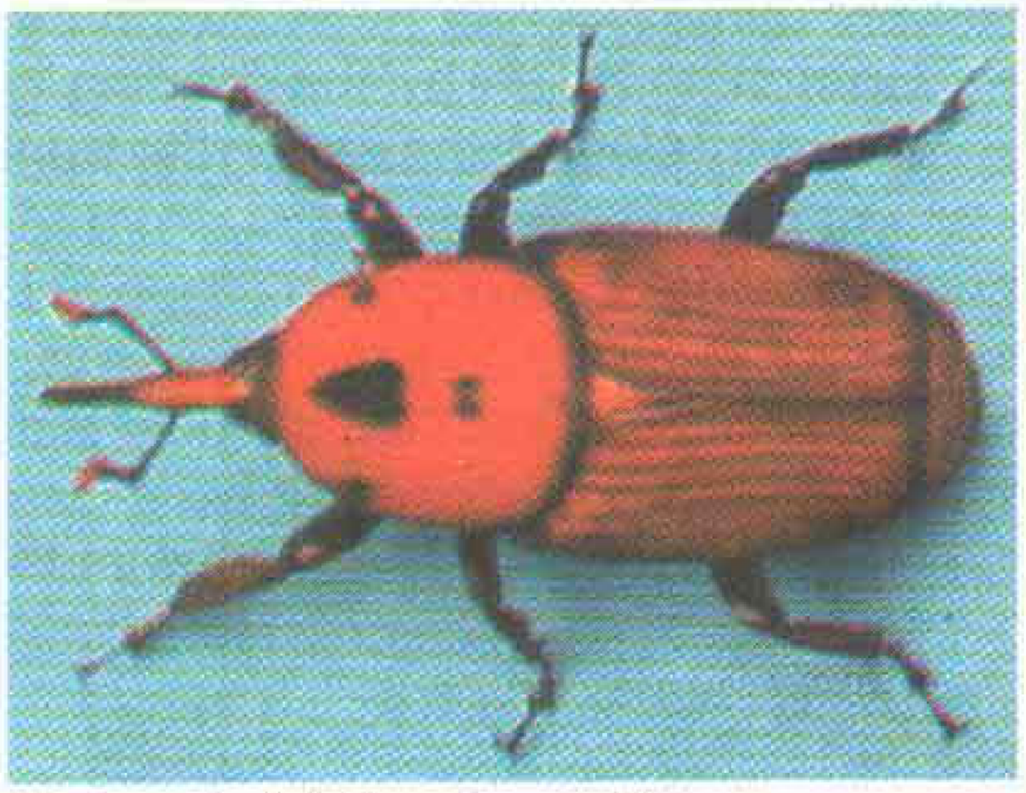
Coconut Research Institute of Sri Lanka

Advisory Circular No B 3

THE RED WEEVIL AND ITS CONTROL

The Red Weevil scientifically known as *Rhynchophorus ferrugineus* is the most serious pest of young coconut palms in the age from 3 to 15 years and is found in all coconut growing areas of the country (Picture 1).



Picture 1: Adult red weevil

Causes for red weevil damage

Red weevil lays eggs only on fresh wounds of young palms. Adult weevil does not make wounds on the palm to lay eggs. They are attracted to fresh wounds caused by animals (Eg: cattle, wild boar), black beatle and human action. Sometime fresh wounds (natural cracks) on the growing stem are caused by enlargement of stem and emergence of aerial roots from the bole region of the stem due to high water table.



Picture 2: A close view of a young palm affected by red weevil

Nature and identification of the damage

The damage is caused by the grubs, which are the larval stage of the red weevil. Due to a large population of grubs feeding on the internal tissues of the trunk, either crown or trunk falls thus killing the palm. Young palms in the age group of 3-15 years are more prone to the attack and palms with large bases are more susceptible as the growth of cracks provide suitable sites for egg laying.

It is difficult to detect the pest damage in the early stage of infestation. In most cases it is detected only after the palm has been severely damaged. However, careful observations may help to detect the presence of the pest. For this purpose it is necessary to clean the crown and the

basal region of the palm. Carefully remove the dried petioles and other debris from the palm without making any fresh injuries. The following signs could be observed if the pest is present in the palm.

- Holes (1.25 - 2 cm) are found in the crown (Picture 2) or the trunk from which chewed up fibers are thrown out. This may be accompanied by the oozing of a brown viscous fluid and the discoloration of the tissues below the holes.

- Crunching noise by the feeding grubs inside the trunk or crown could be heard by placing the ear on the trunk of the palm.

- Yellowing of leaves.

- Withering and slanting of bud (Picture 3).

- Presence of fibrous cocoons at the bases of the petioles and beneath the bark.



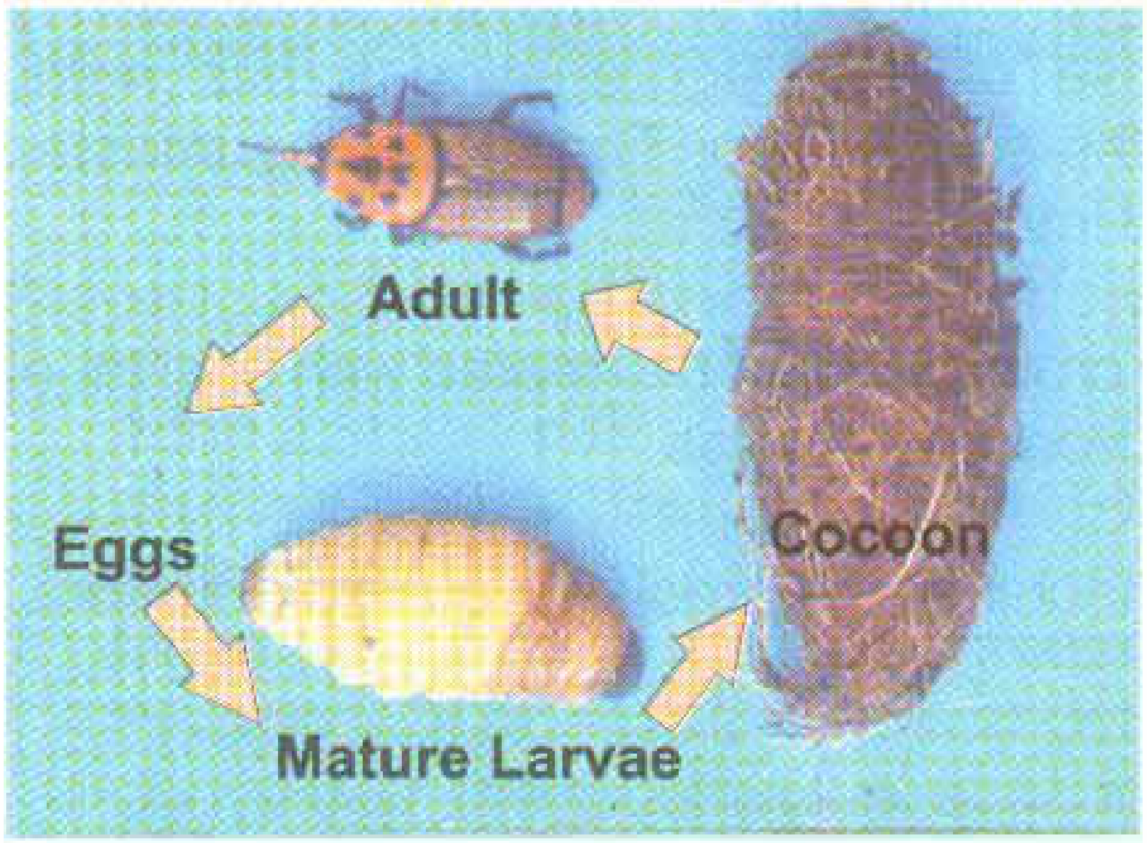
Picture 3: A palm with fallen bud due to red weevil attack



Picture 4: A severely damaged stem of a young palm

Description of the pest

The adult weevil is reddish brown in colour, about 36 mm long and 10 mm in breadth. It is characterized by the presence of long snout on the head. Dark spots are visible on the dorsal middle part of the body. The female red weevil is attracted by the smell of the fermenting sap that emanates from the wounds of the palm. Female weevils lay numerous minute creamy white, oblong eggs in the



Picture 5: Some live specimens of Red weevil's life cycle

fresh wounds on the petioles and trunk. Eggs hatch into grubs (larvae) within 3-5 days which bore in and feed on the fibrous tissues of the trunk and the bud region, thereby killing the palm. The grubs which grow up to 35 mm long can be easily recognized by its brown head and the white leg-less body. After about 55 days it makes a fibrous cocoon near the outer surface of the trunk. Resting stage of the pest inside the cocoon is called pupa. About 20 days later the adult weevil emerges out by cutting a hole either on the trunk or petioles. The emerging males secrete a aggregation pheromone which attracts both male and female weevil. Due to this, the reproductive phase of second generation could also begin in the same infested palm.

Prevention of the pest damage

Difficulties in early detection of the pest may cause a considerable damage to the infested palm. Therefore, emphasis should be placed on preventive aspects of the pest.

- Regular inspection of young palms (between 3 -15 years of age) should be carried out by trained people to detect any fresh wounds and the presence of red weevil at the early stage.

- Coal tar or used engine oil should be applied on fresh wounds on trunks and petioles of all young palms in order to prevent the red weevil from laying eggs.

- Stem bleeding in young palms may be followed by red weevil attack. It is therefore, necessary to treat all bleeding wounds with a recommended fungicide.

- Some times red weevil attacks palms, which have already been attacked by the black beetle. The red weevil lays eggs on the wounds made by black beetle pest.

- An attack on the crown region could be fatal to the palm. It is necessary to prevent the black beetle from breeding in the estates and attacking palms.

- Severely damaged palms, which are beyond recovery, should be cut, spilt and burnt. It is essential to uproot the palms completely in order to destroy all the remaining grubs living in the basal region of the palm.

- Male red weevil emits an aggregation pheromone to signal the other weevils about food/oviposition site. Because of their aggregation habits many generations continuously breed in the same palm. Therefore, the infested palms should be treated or destroyed immediately.

- Any damage caused by cattle on the crown should be treated with coal tar or used engine oil to prevent red weevil from laying eggs.

- Avoid damaging the palms in using machinery and tools in young plantations.

Control

The infested palms should be treated with a systemic insecticide to control the pest. Insecticide treatment is effective during the early stages of the pest attack. If the pest damage has advanced to such an extent to cause withering of bud, treatment will not be effective. Such palms should be cut, split and burnt immediately.

The recommended systemic insecticide is Monocrotophos 60%. The recommended dosage and the trade name of this insecticide are given below.

Insecticide	Dosage	Trade name
1. Monocrotophos	20 ml/palm	Monocrotophos 60%

(A tea spoonful is about 5 ml. Do not use the insecticide measured tea spoon for any domestic purpose)

Monocrotophos 60% is highly toxic and should be handled with extreme care. This chemical is banned in Sri Lanka for other agricultural purposes due to its high toxicity. However, restricted use of insecticide is permitted only for coconut pests. Therefore, it is issued to coconut growers under the supervision of the Coconut Development Officers of the Coconut Cultivation Board

Method of Application

- Trunk injection for palms, which have fully developed trunk.

- Pouring of insecticide to the bud region, for very young palms, which do not have fully developed trunk.

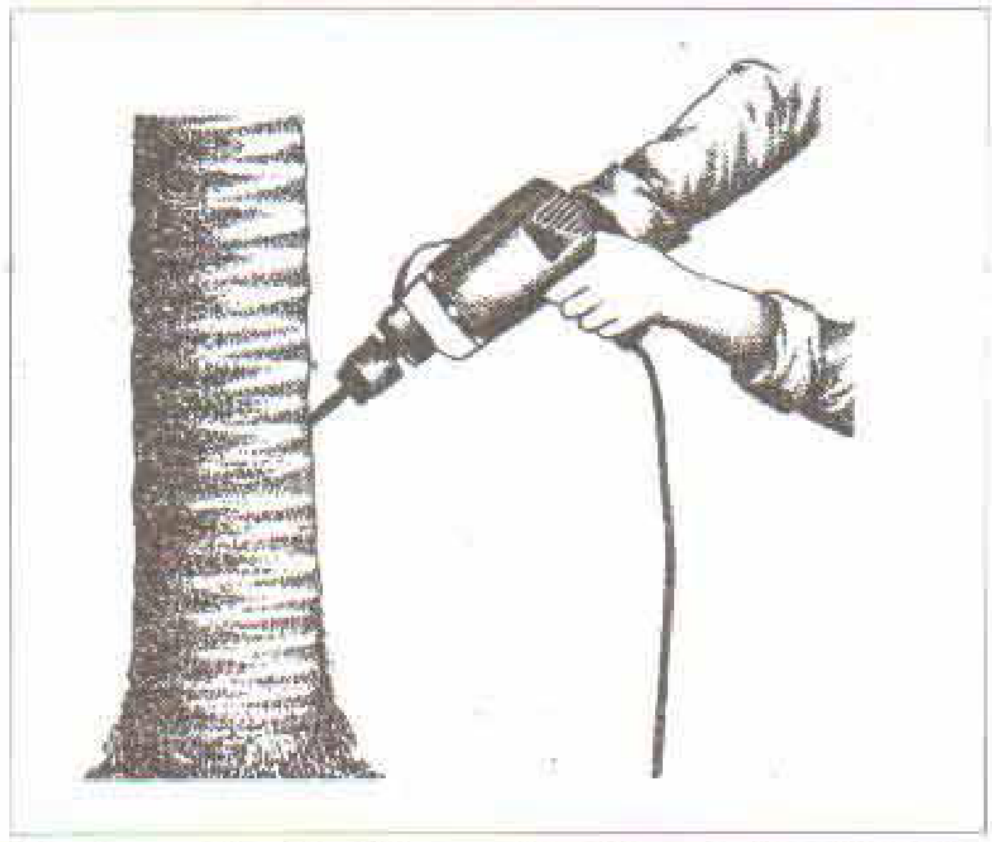
- Root feeding method.

Application of Monocrotophos insecticide

1. Truck injection

Steps to be followed in trunk injection with systemic insecticides

1. If the infested palm bears nuts, pick all mature nuts before insecticide application.
2. Make two holes of 13 mm diameter on the opposite sides in the trunk about 1 m above from the ground level at an angle of 45° and to a depth of 10 cm using an auger or an electric drill (Picture 6).
3. Pour 10 ml of undiluted Monocrotophus 60% using a disposable injection syringe into each hole.
4. Two days after treatment, seal the hole with a cement and sand mixture and later apply coal tar.



Picture 6: Drilling a hole on the trunk

5. Display a warning sign to indicate that the palm is treated with an insecticide.
6. Do not consume the nuts of treated palms until 45 days after injection.

2. New method of trunk injection of a systemic insecticide

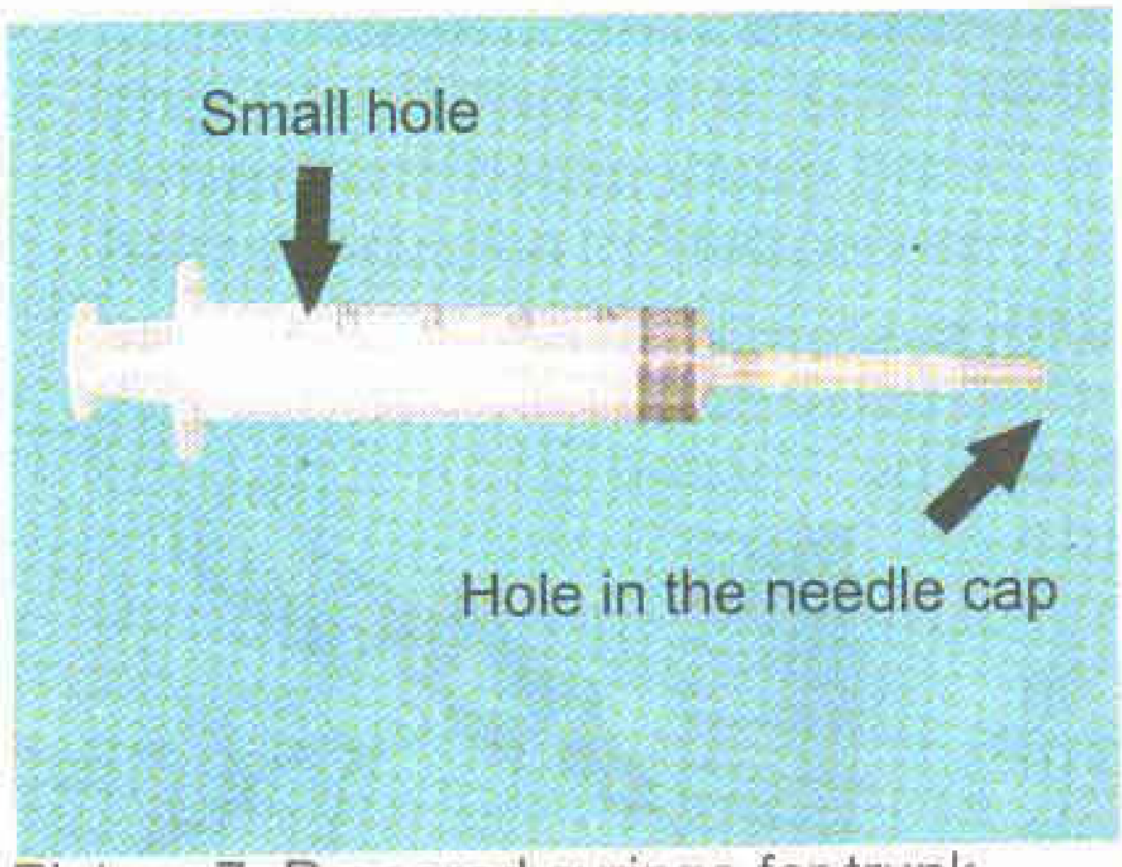
Major disadvantage of the above trunk injection method is making a fairly large two holes on the trunk. These holes would subsequently cause decaying the trunk followed by oozing due to incomplete sealing of holes. The new trunk injection method has been introduced to minimize the damage to the trunk and ensure high efficiency.

Materials needed for trunk injection

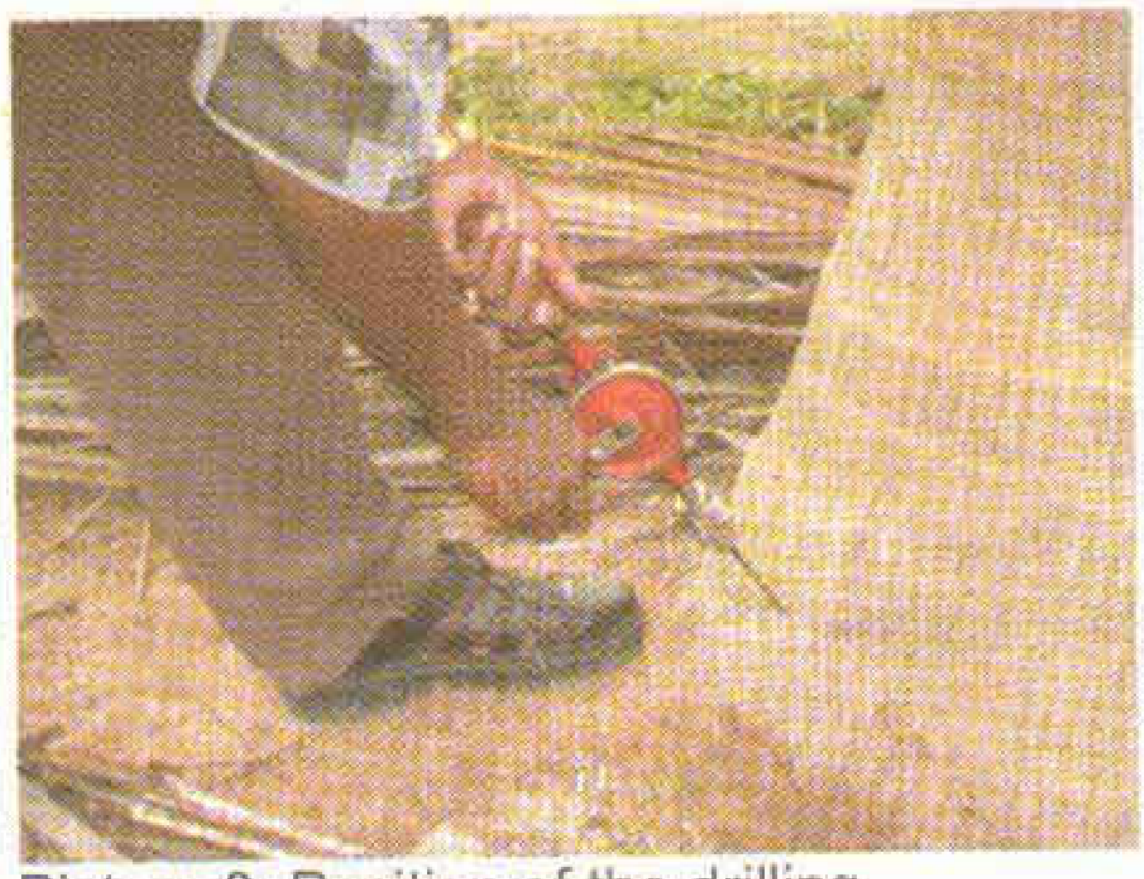
- Drill machine (manually operated or electric) with 4-5 mm diameter drill bit
- Plastic injection syringe with the capacity of 25 ml
- Cap to enclose the injecting needle

Preparation of the syringe

- Purchase a 25 ml disposable injection syringe
- Remove the needle with cap
- Drill a hole at the top end of the syringe using a hand drill with drill bit of 4-5 mm diameter
- Cut the top end of the needle cap to make an opening
- Fix the cap to the syringe end as shown in picture 7. If the cap is loose use a small piece of plastic tube (disposed saline tube) to fix the cap tightly to the syringe
- The syringe is now ready for trunk injection



Picture 7: Prepared syringe for trunk injection



Picture 8: Position of the drilling

Selection of a suitable place for drilling a hole

Select a suitable place with healthy tissues without any wounds. This point should be within 15 cm from the ground level (Picture 8).

Method of drilling and trunk injection

Use a hand drill with a diameter of 4-5 mm. The hole should be 45° in the direction of the base of the trunk. The depth should be 2.5 - 3.75 cm. Drilling should be repeated to get a fiber free clean hole. Suck 20 ml monocrotophos 60% into the syringe. Insert the open end of the needle cap into the hole and fix syringe tightly (Picture 9). Pull the piston of the syringe slowly above the hole made. Check for any leakage.



Picture 9: Fixing the syringe

The diameter of the hole should be small in order to prevent the spilling of insecticide from the hole. The installed syringe will have to be removed after the palm absorbs entire insecticide. The absorption of the insecticide normally takes two days. The treated hole should be sealed with ordinary candle wax.

3. For young palms which do not have a fully formed trunk

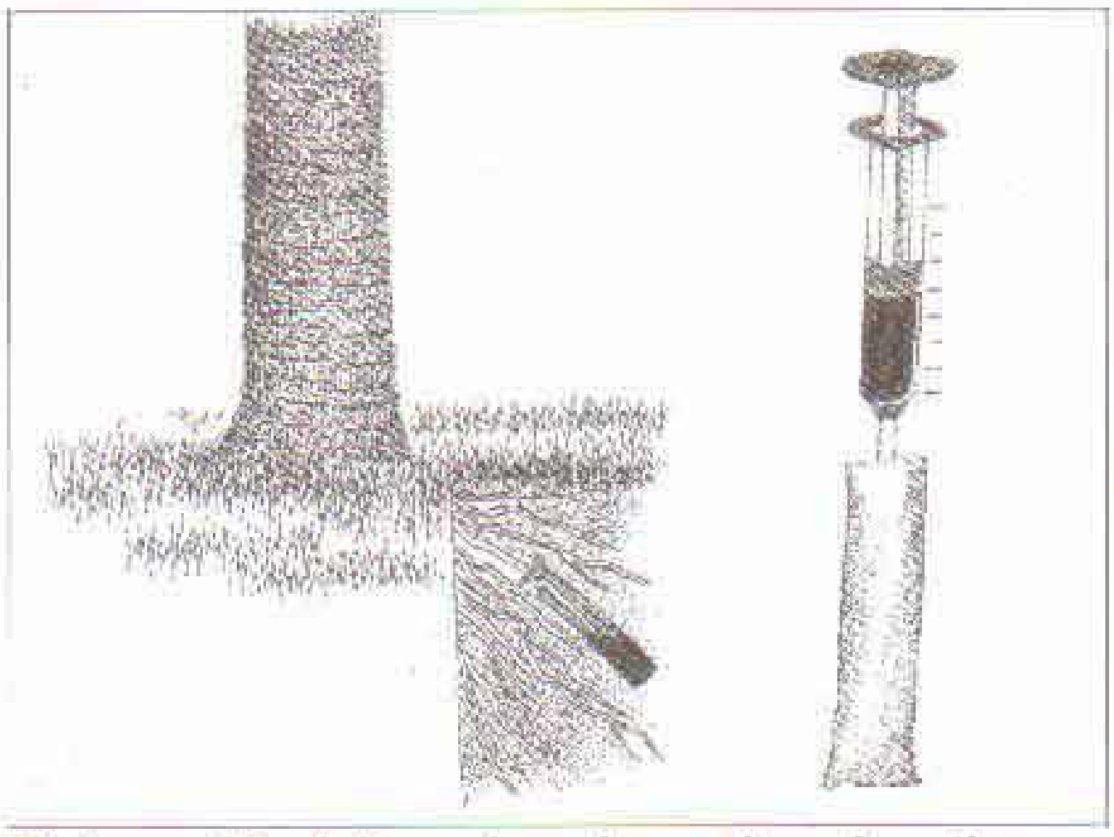
Mix 10 ml of Monocrotophus 60% with 500 ml of water and pour into the crown to wet the bud region thoroughly.

4. Root feeding of monocrotophos

If the infested palm bears nuts, harvest all mature nuts before root feeding. Carefully expose a healthy new root with a thickness of a pencil. The roots in light brown colour are found to be effective for root feeding treatment. The dosage of

Monocrotophos 60% (20 ml / palm) should be mixed with an equal amount of water. Then the mixture should be transferred to a suitable container (long and narrow bag prepared with thick polythene, narrow glass tube or a small narrow bottle) and insert the cut edge of root to the bottom of insecticide liquid (Picture 10 and 11). Please note that the selected root should be cut carefully with a sharp knife just above the root tip.

On inspection on the following day, if the insecticide has not been absorbed at all, transfer the container to a new root. The treated area should be well protected from human or animal intervention and prevent from other mechanical damages until the insecticide absorption is completed. The nuts of treated palms should not be consumed for two months from the time of treatment.



Picture 10: A line drawing showing the method of root feeding



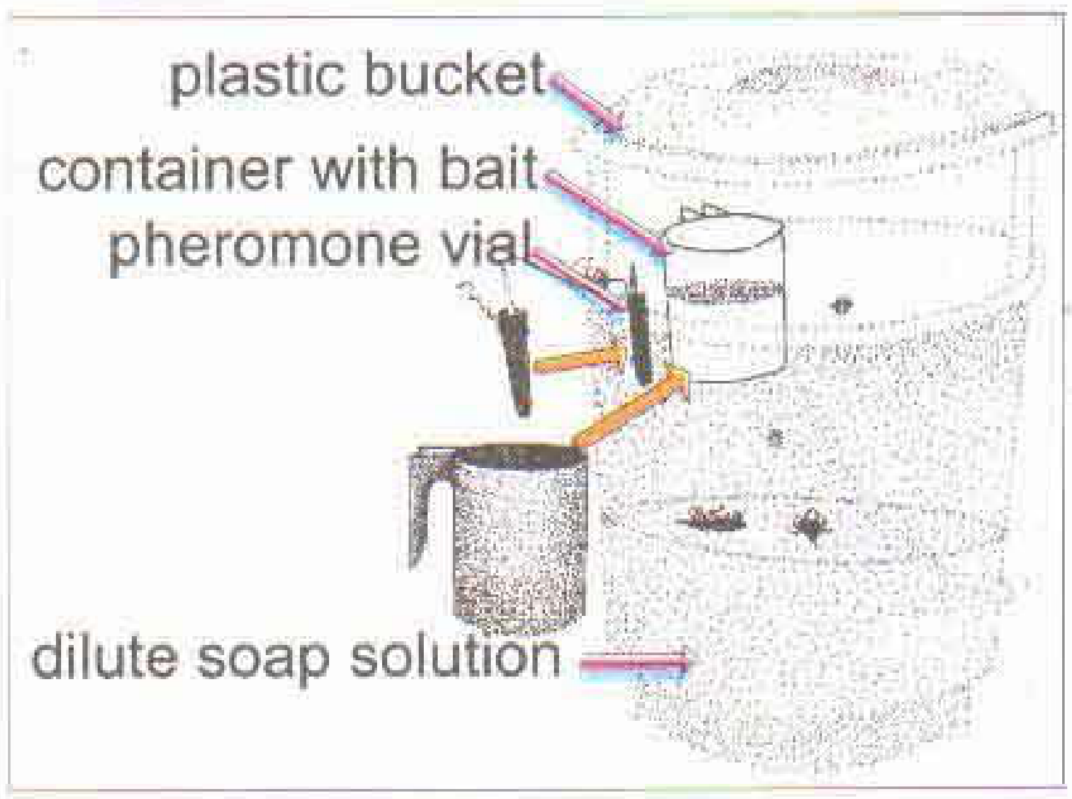
Picture 11: A treated root

Management of red weevil using pheromone trap

The pheromone trap is recommended for estates with more than 5 acres extent. If the pest population is very high and above the economic thresh hold level, then we can use pheromone traps to control the population.

Installation of pheromone trap

Each trap is consisting of a plastic bucket, pheromone vial and a small container with bait. Pour about 3" (7.5 cm) depth of dilute soap solution and keep the bait container at the mouth of the bucket. You can use any one of the baits such as sugar solution, toddy or yeast sugar solution. Carefully break the neck of the pheromone-containing vial, and tie it to the handle of the bucket at the mouth level of the bucket. For the successful control of the pest 5-6 traps/hectare is recommended (Picture 12).



Picture 12: Pheromone trap

If it is old plantation the traps could be tied on the trunks at convenient height. The traps should be installed at different places of the estate to provide maximum control of the pest. If the plantation is young use sticks as stakes to tie the pheromone traps (Picture 13).



Picture 13: Method of installing the trap

Method of installation of the trap in the field

Every two weeks the traps should be cleaned and refilled with fresh soap solution and bait. If the population density of the pest is high repeat the pheromone control method every 5-6 months. If the population is under control, then we can stop the pheromone application.

Note 1: Preparation of yeast sugar solution

Materials	**quantity**
Sugar	150 (g)
Yeast	2 (g)
Water	1 liter

Prepare a homogenous solution of sugar/yeast mixture. Use 100 ml solution/trap as bait.

For further detail contact Director, Coconut Research Institute, Lunuwila or Coconut Development Officer of your area.

Coconut Research Institute of Sri Lanka, Lunuwila, December - 2006